Wrong and Useful Models

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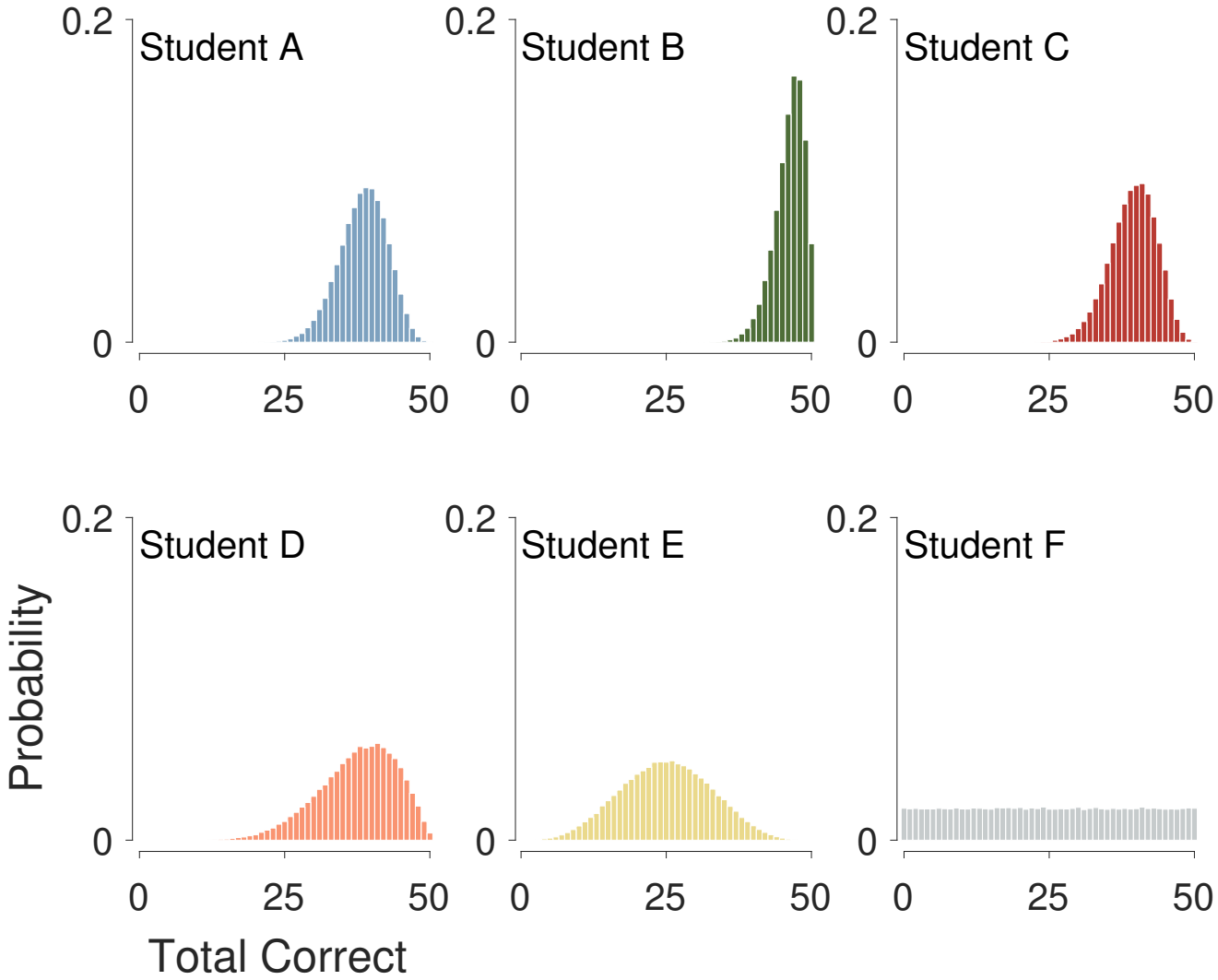
Student Grades

- A class of students are taking a fifty-item exam
  - students A–C have completed the exam
  - students E and F have answered the first 10 questions
  - student F has not started
- The success rate model can be applied by treating correct answers to questions as successes

Student	Correct	Completed	Percentage
A	39	50	78%
B	47	50	94%
C	40	50	80%
D	8	10	80%
E	5	10	50%
F	0	0	–

Success Count Posterior Predictions

- The posterior predictions are for the number of correct questions $\hat{k}$ once the exam is completed with $\hat{n} = 50$



Is the Model Wrong?

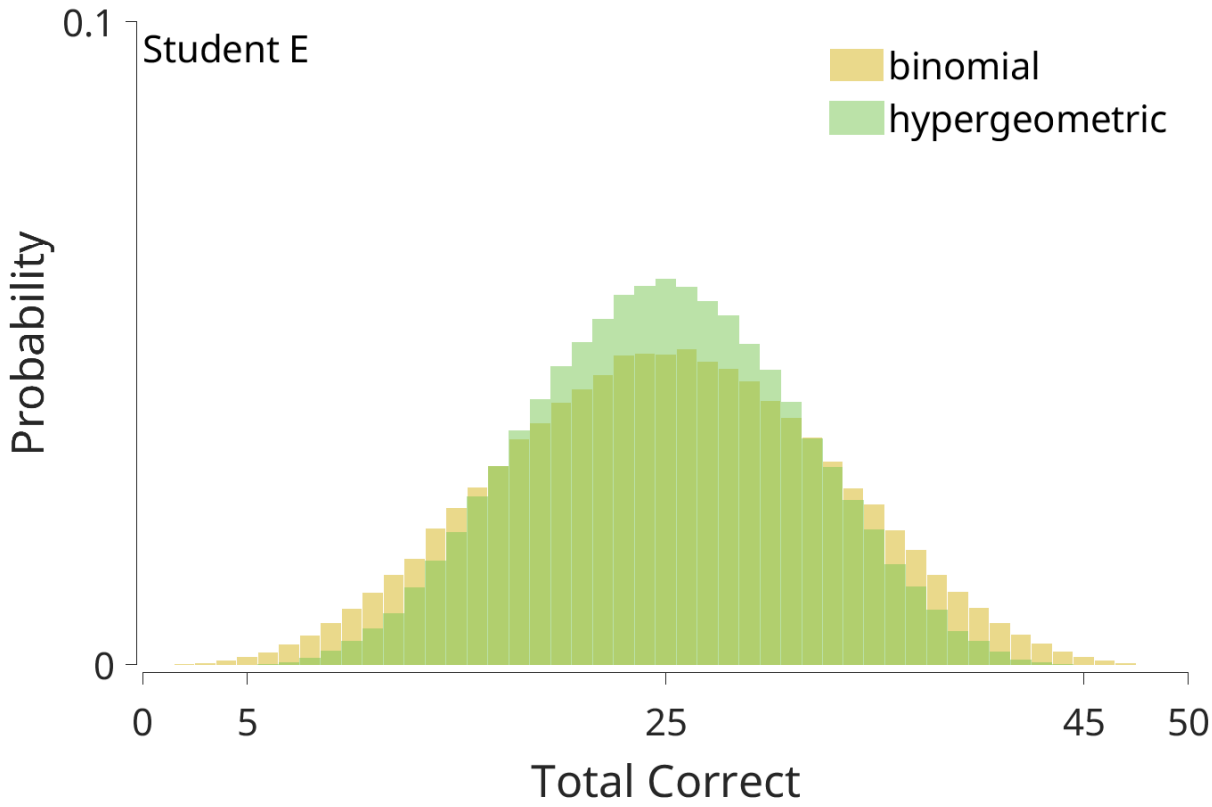
- If we think of the posterior predictive distribution as a prediction for what the same student would do if they took the exam again, the results make sense
  - student A might not get exactly 39 out of 50 again and their posterior predictive distribution quantifies the probability of the new outcome
- If we think of the posterior predictive distribution as a prediction for this exam the results make less sense
  - we know the final scores for students A, B, and C exactly, and there is no uncertainty to be distributed
- The results are also not right for predictions about the final scores on this exam for students D and E
  - student D is constrained to score between 8 and 48
  - student E is constrained to score between 5 and 45

The Core Problem

- The problem is that the success rate model is "wrong" because it uses the $\mathrm{binomial}$ distribution, which samples the questions with replacement
- Under the "this exam" interpretation
  - all of the questions are observed for students A, B, and C and there is nothing to infer
  - the first ten questions are observed for students E and F, and should not be replaced

A Better Model

- An alternative model uses the $hypergeometric$ distribution, which samples without replacement, and meets the constraints
  - the posterior predictive distributions for student E based on both models are shown
  - the differences matter if the research question involves extreme predictions



But the Model is Still Wrong

- The hypergeometric model is probably also wrong in some ways, including in terms of assumptions shared with the binomial model
  - e.g., every question is unlikely to be equally difficult
  - e.g., students' success rates probably vary with motivation or fatigue
- Good scientific models should strive to be useful in the context in which they applied for the scientific inferences they need to answer

